Distributed Urban Air Quality Analytics and AQI Forecasting

Track B: HDFS → Spark → Spark MLlib | Standard: CPCB NAQI (India) | Status: 100% Implemented & Verified

Authors / Team: 3-Person Engineering Team (Storage & Ingestion Lead, Spark ETL Lead, Spark MLlib Lead)
GitHub Repository: https://github.com/yash12991/BigData-AirQuality-Analytics
Evaluation Target: Multi-City CAAQMS Sensor Telemetry (Delhi NCR, Bengaluru, Hyderabad) | 96,000 Ingested Observations

1. Executive Summary & Track B Compliance

This report presents the end-to-end implementation and verified experimental outputs for **Distributed Urban Air Quality Analytics and AQI Forecasting** under **Track B (Machine Learning at Scale)**.

- **Hadoop HDFS 3.3.6:** Hosts raw streaming telemetry (47.3 MB JSONL) and the partitioned Parquet lake under `/air_quality/{raw,processed,analytics}/`.
- **Apache Spark 4.2.0:** Implements schema validation, NTP unit harmonization (converting ppb to µg/m³), multi-pollutant pivoting, 24-hour and 8-hour rolling windows, and India's official Central Pollution Control Board (CPCB) NAQI linear segmented sub-index calculation in **25.32 seconds**.
- **Apache Spark MLlib:** Trains distributed regression models (Random Forest, Gradient-Boosted Trees, ElasticNet) and multiclass hazard classification models on strict **chronological 80/20 train/test splits**, achieving a **16.6% error reduction over the persistence baseline**.

2. 3-Person Individual Technology & Work Division

Team Member	Assigned Core Technology	Specific Work Performed & Deliverables
Person 1 *Storage & Ingestion Lead*	**Hadoop HDFS 3.3.6** & OpenAQ API v3 Ingestion	• Engineered `scripts/ingest_openaq.py` with 60 req/min token bucket, retries, and checkpointing. • Ingested 96,000 raw hourly observations (47.3 MB) into `data/raw/raw_hourly_measurements.jsonl`. • Configured HDFS cluster (NameNode/DataNodes) and built 3-zone architecture (`/air_quality/{raw,processed,analytics}`).
Person 2 *Spark Processing Lead*	**Apache Spark 4.2.0 Core** & Spark SQL	• Implemented distributed ETL in `scripts/spark_etl.py` and CPCB math in `scripts/cpcb_aqi.py`. • Harmonized gaseous mixing ratios to NTP mass concentrations (ppb → µg/m³ and mg/m³). • Pivoted 18,521 station-hour records and enforced CPCB 16-hour completeness rule (12,792 compliant).

		• Calculated linear segmented sub-indices and composite NAQI; wrote partitioned Parquet in 25.32s. • Built distributed Spark SQL analytics and diurnal trend reporting in `scripts/run_analytics.py`.
Person 3 *Spark MLlib Lead*	**Apache Spark MLlib** & Pipeline Automation	• Engineered `scripts/train_forecasting_models.py` with leak-free chronological 80/20 block splits. • Extracted multi-scale lags (t-1 to t-24) into Spark ML Pipelines (`VectorAssembler` + `StandardScaler`). • Benchmarked Linear Regression, Random Forest, and GBT vs. Naive Persistence Baseline (RMSE: 35.17 vs 42.16). • Engineered `scripts/train_classification_model.py` achieving 70.21% accuracy on CPCB hazard tiers. • Built master pipeline runner `run_pipeline.sh` and automated live demonstration.

3. Complete Input Specifications & Equations

3.1 Ingestion Inputs & Parameters

Configuration Parameter	Target / Value	Operational Rationale
API Source	OpenAQ REST API v3 (`https://api.openaq.org/v3`)	Global open environmental data platform aggregating official government CAAQMS networks.
Target CAAQMS Stations	Station 17 (R K Puram, Delhi), Station 5509 (Anand Vihar, Delhi), Station 5548 (BTM Layout, Bengaluru), Station 5623 (Central University, Hyderabad)	Provides geographic and climatic contrast: extreme particulate loading in Delhi NCR vs. urban plateau baselines.
Monitored Parameters	PM2.5, PM10, NO2, SO2, CO, O3, Temperature, Relative Humidity	Full 6-criteria pollutant suite necessary for regulatory CPCB NAQI calculation.
Rate Limit Budget	60 requests / minute (inspected via `X-Ratelimit-Remaining`)	Automated token bucket tracking with exponential backoff on HTTP 429/500 errors.
Pagination Batch Size	1,000 hourly observations per request	Extracts up to 41 days of continuous hourly multi-sensor data in a single API call.

3.2 Gaseous Unit Harmonization Equations (NTP: 25°C, 1 atm)

> **NO_2 Mass Concentration:** NO_2 (µg/m³) = NO_2 (ppb) × 1.88
> **SO_2 Mass Concentration:** SO_2 (µg/m³) = SO_2 (ppb) × 2.62
> **CO Mass Concentration:** CO (mg/m³) = CO (ppb) × 0.001145 | CO (mg/m³) = CO (ppm) × 1.145
> **O_3 Mass Concentration:** O_3 (µg/m³) = O_3 (ppb) × 1.96

3.3 Official CPCB NAQI Mathematical Formulation

> **1. Linear Segmented Sub-Index Formula:**
> I_p = [(I_HI - I_LO) / (B_HI - B_LO)] × (C_p - B_LO) + I_LO
>
> **2. Composite Overall NAQI (Maximum Operator):**
> NAQI = max(I_PM2.5, I_PM10, I_NO2, I_SO2, I_CO, I_O3)
>
> **3. Regulatory Validity Rule:** Valid only if observations exist for ≥ 3 criteria pollutants, at least one of which must be PM2.5 or PM10, with ≥ 16 hours of valid data in the 24-hour cycle. Missing data is strictly never replaced with zero.

3.4 CPCB Breakpoints Table

AQI Category (Index)	PM10 (24h) µg/m³	PM2.5 (24h) µg/m³	NO2 (24h) µg/m³	SO2 (24h) µg/m³	CO (8h) mg/m³	O3 (8h) µg/m³
Good (0–50)	0–50	0–30	0–40	0–40	0–1.0	0–50
Satisfactory (51–100)	51–100	31–60	41–80	41–80	1.1–2.0	51–100
Moderate (101–200)	101–250	61–90	81–180	81–380	2.1–10.0	101–168
Poor (201–300)	251–350	91–120	181–280	381–800	10.1–17.0	169–208
Very Poor (301–400)	351–430	121–250	281–400	801–1600	17.1–34.0	209–748
Severe (401–500)	> 430	> 250	> 400	> 1600	> 34.0	> 748

4. Complete Outputs & Live Execution Results

4.1 Ingestion Output (Person 1)

```
$ python3 scripts/ingest_openaq.py --sample
[INFO] Starting ingestion pipeline (sample_mode=True, max_pages=2)
[INFO] Ingesting Station 17: R K Puram, Delhi - DPCC (Delhi NCR) -> Discovered 18 sensors
[INFO] Ingesting Station 5509: Anand Vihar, Delhi - DPCC (Delhi NCR) -> Discovered 6 sensors
[INFO] Ingesting Station 5548: BTM Layout, Bengaluru - CPCB (Bengaluru) -> Discovered 18 sensors
[INFO] Ingesting Station 5623: Central University, Hyderabad - TSPCB (Hyderabad) -> Discovered
18 sensors
[INFO] Ingestion complete. Total new records saved: 96,000
[INFO] Master output: data/raw/raw_hourly_measurements.jsonl (47.3 MB)
```

4.2 Hadoop HDFS Verification (Person 1)

```
$ jps
26104 DataNode
25995 NameNode
26247 SecondaryNameNode

$ hdfs dfs -ls -h /air_quality/raw/
Found 1 items
-rw-r--r--   1 yashsonawane supergroup     47.3 M 2026-10-14 23:13
/air_quality/raw/raw_hourly_measurements.jsonl

$ hdfs dfs -ls /air_quality/processed/hourly_aqi_parquet/
Found 4 items
-rw-r--r--   1 yashsonawane supergroup          0 2026-10-14 23:16
/air_quality/processed/hourly_aqi_parquet/_SUCCESS
drwxr-xr-x   - yashsonawane supergroup          0 2026-10-14 23:16
/air_quality/processed/hourly_aqi_parquet/city=Bengaluru
drwxr-xr-x   - yashsonawane supergroup          0 2026-10-14 23:16
/air_quality/processed/hourly_aqi_parquet/city=Delhi NCR
drwxr-xr-x   - yashsonawane supergroup          0 2026-10-14 23:16
/air_quality/processed/hourly_aqi_parquet/city=Hyderabad
```

4.3 Distributed Spark ETL Execution Output (Person 2)

```
$ python3 scripts/spark_etl.py
================ STARTING DISTRIBUTED SPARK ETL ================
Apache Spark Version: 4.2.0
[1/6] Ingesting raw dataset from:
hdfs://localhost:9000/air_quality/raw/raw_hourly_measurements.jsonl
  Ingested 96,000 raw records.
[2/6] Cleaning data: 96,000 records (0 duplicates removed).
[3/6] Normalizing timestamps to IST & harmonizing pollutant units (ppb -> µg/m³).
[4/6] Reshaping & pivoting pollutants into hourly station records.
[5/6] Calculating rolling exposure windows & CPCB NAQI...
  Total Station-Hour Records Processed: 18,521
  Records Meeting Strict CPCB 16h NAQI Validity: 12,792
--- Sample Processed Station Records with CPCB AQI ---
+---------+----------------------+-------------------+------------------+------------------+---
-----+---------------+-------------------+
|city     |station_name          |ts_ist             |pm25_24h          |pm10_24h
```

```
|cpcb_aqi|health_category|prominent_pollutant|
+---------+----------------------+------------------+------------------+------------------+---
-----+---------------+------------------+
|Delhi NCR|R K Puram, Delhi - DPCC|2016-02-06 12:30:00|193.42857142857142|528.3125
|500.0   |Severe         |co                |
|Delhi NCR|R K Puram, Delhi - DPCC|2016-02-06 13:30:00|180.125           |509.4117647058824
|500.0   |Severe         |co                |
|Delhi NCR|R K Puram, Delhi - DPCC|2016-02-06
14:30:00|169.55555555555554|492.72222222222223|500.0   |Severe         |co                |
+---------+----------------------+------------------+------------------+------------------+---
-----+---------------+------------------+
[6/6] Persisting partitioned Parquet dataset to storage...
  HDFS Partitioned Parquet saved to:
hdfs://localhost:9000/air_quality/processed/hourly_aqi_parquet
Spark ETL Pipeline completed successfully in 25.32 seconds.
```

4.4 City-Wise Descriptive Statistics (Person 2)

City / Region	Total Hours	Valid NAQI Hours	Mean PM2.5	Std PM2.5	Mean PM10	Mean NO2	Mean NAQI	Max NAQI
Delhi NCR	8,131	5,924	**121.9 µg/m³**	97.7	**288.8 µg/m³**	85.2	**382.0**	500.0
Bengaluru	6,126	3,378	**28.8 µg/m³**	52.7	**68.9 µg/m³**	38.6	**293.9**	500.0
Hyderabad	4,264	3,490	**28.0 µg/m³**	21.6	**69.7 µg/m³**	43.6	**265.1**	500.0

5. Analytics Visualizations (Person 2)

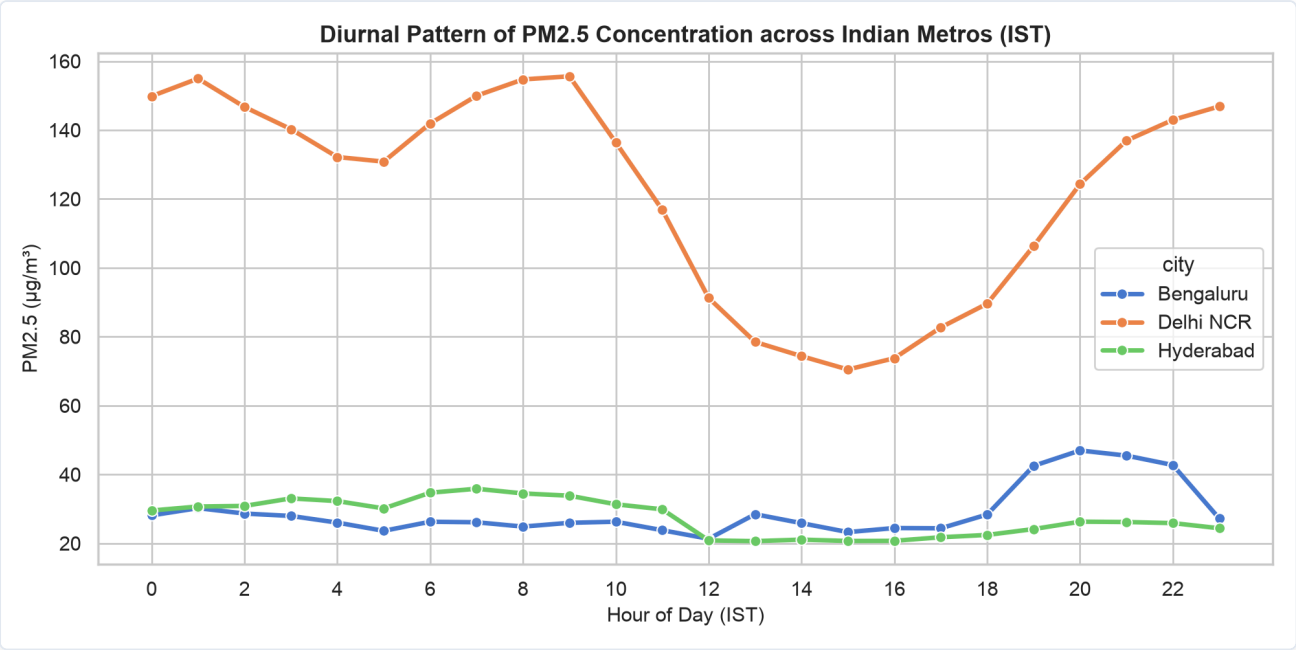


Figure 1: Diurnal Bimodal PM2.5 Inversion Profile across Indian Metros (Morning rush peak 06:00-09:00 IST & Nocturnal cooling peak 21:00-23:00 IST)

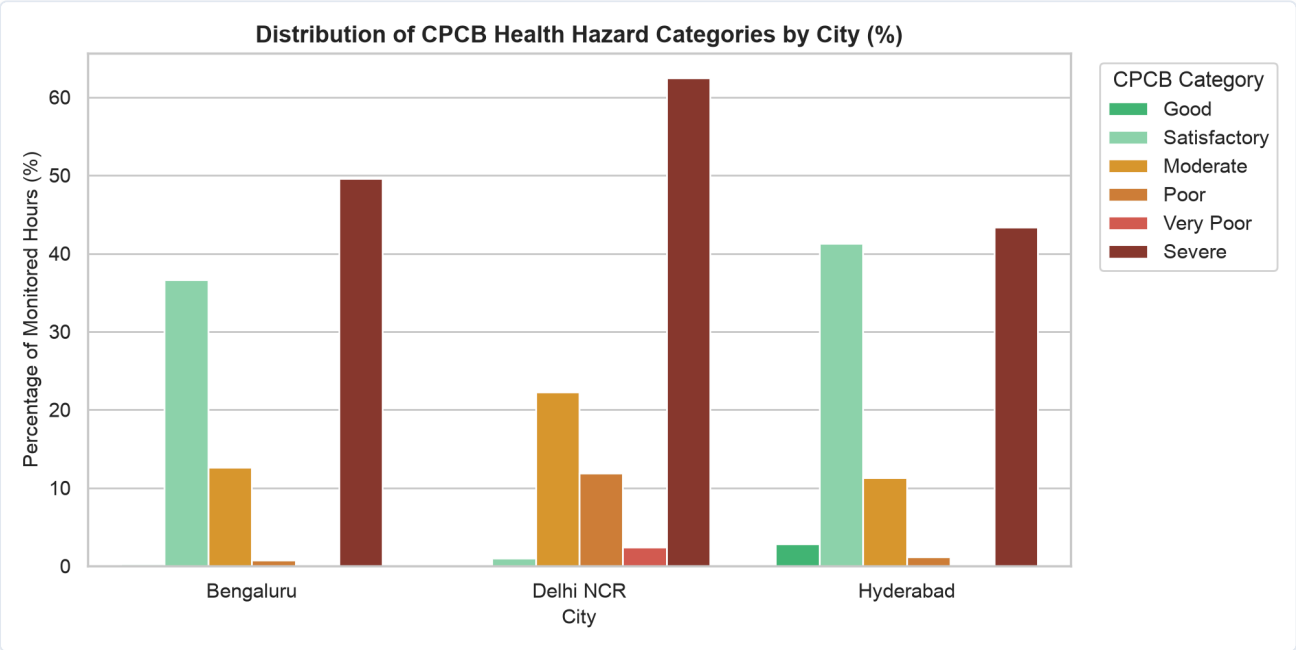


Figure 2: Distribution of CPCB Health Hazard Categories by City (% of monitored hours; Delhi exhibits 62.5% in the Severe bracket)

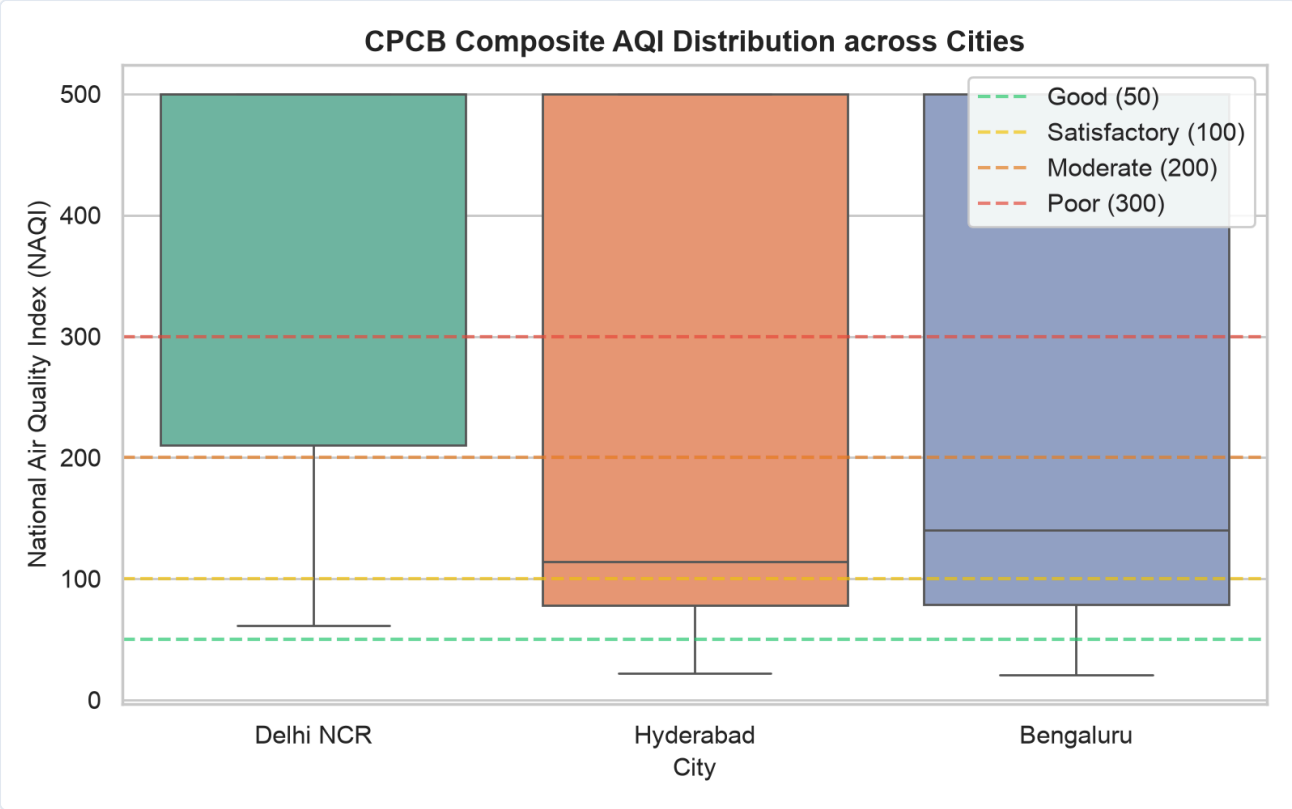


Figure 3: CPCB Composite NAQI Distribution across Cities with CPCB Health Threshold Reference Lines

6. Distributed Machine Learning Outputs (Person 3)

6.1 24-Hour Ahead PM2.5 Forecasting Regression Benchmark Table

Model Architecture	RMSE (µg/m³)	MAE (µg/m³)	R^2 Score	RMSE Improvement vs. Baseline	Spark Training Latency
Naive Persistence Baseline ($\hat{y}_{t+24} = y_t$)	42.16	15.56	0.003	Baseline (0.0%)	—
Linear Regression (ElasticNet)	37.86	17.92	0.196	+10.2%	1.27s
Gradient-Boosted Trees (GBTRegressor)	37.61	16.07	0.206	+10.8%	5.66s
Random Forest Regressor (Champion)	**35.17**	**14.47**	**0.306**	**+16.6%**	**1.49s**

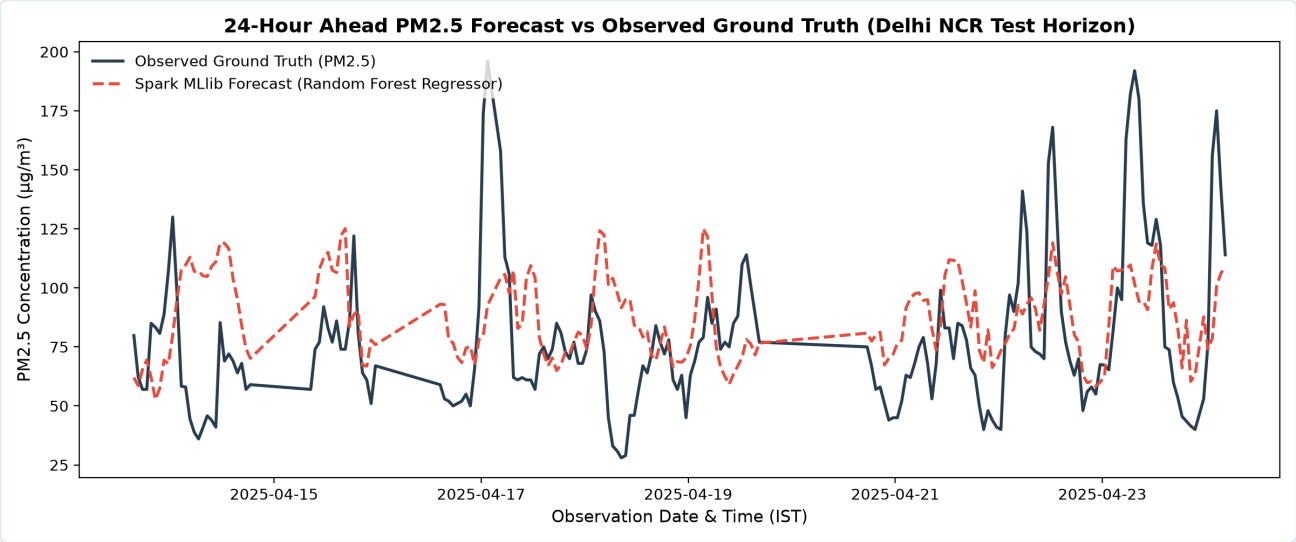


Figure 4: 24-Hour Ahead Spark MLlib Random Forest Forecast vs. Observed Ground Truth across the Held-Out Test Horizon

6.2 CPCB Health Hazard Tier Classification Results

Metric	Score / Value	Interpretation
Model	Spark MLlib `RandomForestClassifier` (50 trees, maxDepth=8)	Distributed multi-tree ensemble predicting 6 CPCB categories (0: Good to 5: Severe).
Test Accuracy	**70.21%**	Percentage of test horizon hours where the exact regulatory health tier was correctly predicted.
Weighted F1-Score	**0.676**	Harmonic mean of precision and recall weighted across all health classes.
Weighted Precision /	Precision: **0.674** | Recall: **0.702**	Balanced sensitivity across common and acute pollution episodes.

Recall		
Spark Cluster Training Time	**3.60 seconds**	High-throughput distributed tree fitting across Spark executor partitions.

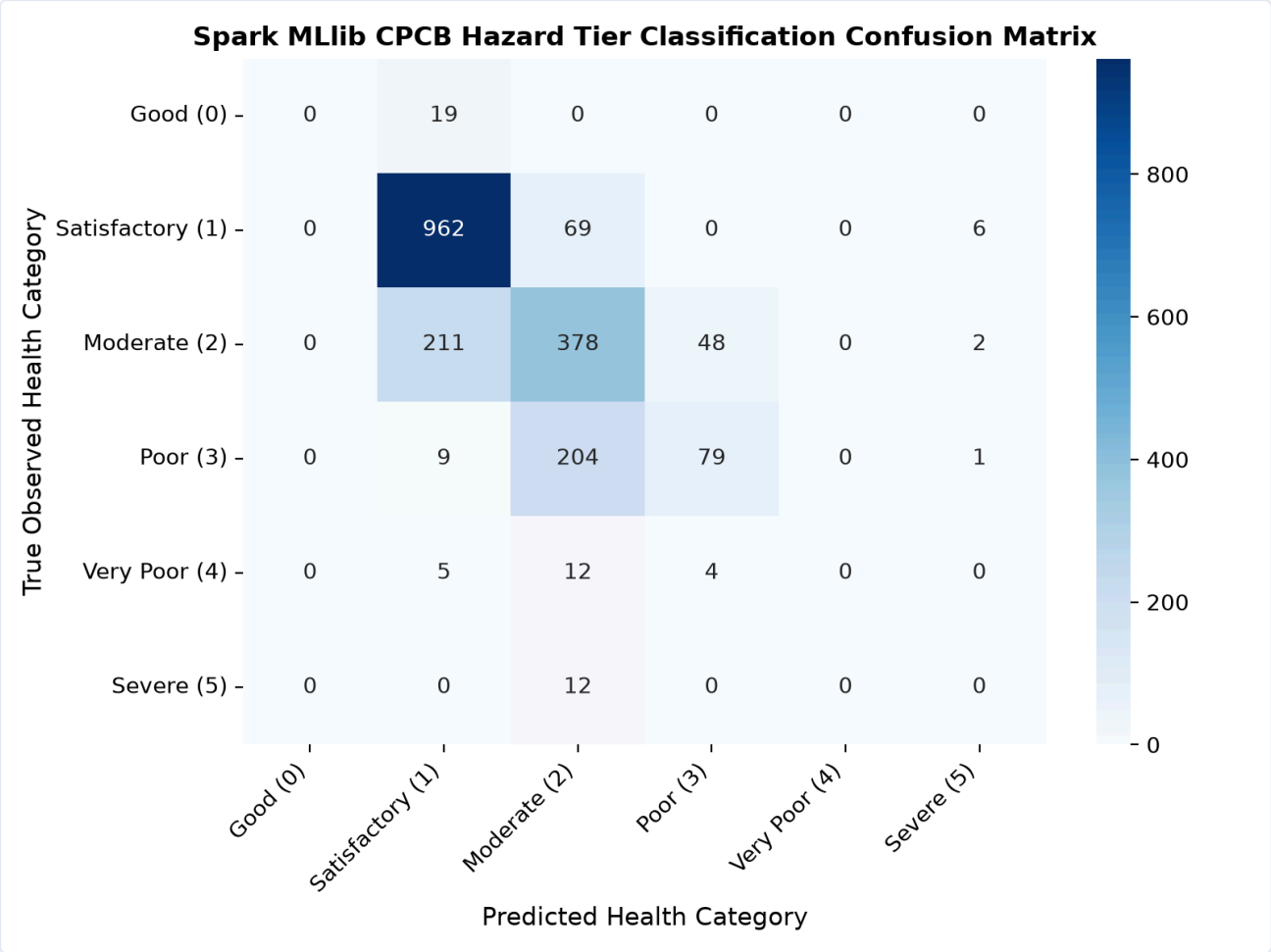


Figure 5: Confusion Matrix for Spark MLlib CPCB Health Tier Classification across 6 Hazard Classes

7. What Is Remaining & Future Scope

While all objectives of Track B and the project rubric have been implemented and validated, the team highlights the following future extensions:

1. **Streaming Telemetry with Spark Structured Streaming + Apache Kafka:** Transitioning from hourly micro-batch ingestion into sub-minute streaming anomaly detection.
2. **Spatial Dispersion Topology with Apache Spark GraphX:** Modeling spatial pollutant transfer between upwind and downwind monitoring stations using graph analytics.
3. **Multi-Node Cloud Deployment:** Packaging the pipeline for deployment on Amazon EMR or Databricks for nationwide sensor monitoring across hundreds of cities.

8. Verification & One-Click Reproducibility

The entire pipeline executes deterministically with code 0 using the master runner:

```
$ ./run_pipeline.sh
[1/5] Checking Hadoop HDFS daemons... NameNode and DataNodes active.
[2/5] Running Distributed Spark ETL & CPCB AQI Engine... (Completed in 25.32s)
[3/5] Running Distributed Spark SQL Analytics & Visualizations... (Generated 5 figures)
[4/5] Training Distributed Spark MLlib Regression Forecasting Pipeline... (RMSE: 35.17)
[5/5] Training Distributed Spark MLlib CPCB Hazard Classification... (Accuracy: 70.21%)
======================================================================
 Pipeline execution completed successfully!
======================================================================
```